

# CredNest AI: An Intelligent Financial Management Platform with Multi-Agent AI Architecture for the Indian Financial Ecosystem

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## ABSTRACT

The rapid digitization of financial services in India has created an urgent need for intelligent platforms that can democratize access to personalized financial guidance. This paper presents CredNest AI, a comprehensive web-based financial management system that leverages a multi-agent artificial intelligence architecture to provide domain-specific advisory services across budgeting, investments, taxation, insurance, loans, and government welfare schemes. The platform integrates real-time market data aggregation with context-aware conversational AI agents powered by large language models (LLMs), enabling users to receive actionable financial recommendations grounded in current market conditions. We detail the system architecture comprising a React-based progressive frontend, serverless edge functions for AI orchestration, and a PostgreSQL-backed data layer with row-level security. Our implementation demonstrates the feasibility of building production-grade, multi-domain financial advisory systems using modern serverless architectures while maintaining data privacy and regulatory compliance relevant to the Indian financial regulatory framework. Preliminary user evaluation indicates significant improvement in financial literacy metrics and decision-making confidence among participants from diverse socioeconomic backgrounds.

**Keywords**—Financial Technology, Multi-Agent AI Systems, Large Language Models, Personal Finance Management, Indian Financial Ecosystem, Serverless Architecture, Real-time Data Aggregation

## I. INTRODUCTION

India's financial landscape has undergone a transformative shift over the past decade, driven by the proliferation of digital payment systems such as UPI, the expansion of mutual fund investments through platforms like AMFI, and governmental initiatives promoting financial inclusion through schemes such as Jan Dhan Yojana and Atal Pension Yojana. Despite these advancements, a significant portion of the population—

particularly in semi-urban and rural demographics—continues to face challenges in navigating the complex matrix of financial products, taxation regulations, and government welfare programs. According to the National Centre for Financial Education (NCFE) survey of 2024, only 27% of Indian adults demonstrate adequate financial literacy, indicating a substantial gap between the availability of financial services and the capacity of individuals to leverage them effectively.

Existing financial management applications predominantly fall into two categories: passive tracking tools that merely aggregate transaction data without providing actionable intelligence, and premium advisory services that remain economically inaccessible to the average user. The emergence of large language models (LLMs) presents an unprecedented opportunity to bridge this divide by enabling natural language-driven financial advisory services that can contextualize recommendations based on individual financial profiles, current market conditions, and regulatory frameworks.

This paper introduces CredNest AI, a full-stack financial management platform that addresses these challenges through the following principal contributions:

- (1) A multi-agent AI architecture employing domain-specialized conversational agents for finance, taxation, and government schemes, each equipped with contextual knowledge bases and real-time computational tools.
- (2) A real-time market data aggregation pipeline that synthesizes information from banking institutions, insurance providers, and investment fund houses to provide current comparative analysis.
- (3) A comprehensive tax planning and ITR filing assistant that implements both Old and New Indian tax regimes with section-wise deduction optimization.
- (4) A financial health scoring algorithm that computes a holistic wellness metric by analyzing income patterns, expenditure behavior, savings trajectories, and debt obligations.
- (5) A privacy-preserving architecture leveraging row-level security (RLS) policies ensuring that user financial data remains isolated and protected at the database layer.

## II. RELATED WORK

### A. Personal Finance Management Systems

Prior work in personal finance management (PFM) systems has explored various approaches to automated financial tracking and advisory. Mint (Intuit) and YNAB pioneered the category by providing transaction categorization and budget

tracking capabilities. However, these platforms primarily operate as passive data aggregators without intelligent advisory capabilities. Koo et al. (2023) proposed an AI-enhanced budgeting system using rule-based classification, but their approach lacked the contextual understanding necessary for personalized recommendations in diverse regulatory environments such as India's dual tax regime system.

### **B. AI-Driven Financial Advisory**

The application of conversational AI in financial services has been explored by several research groups. Loukas et al. (2023) demonstrated the use of fine-tuned language models for investment sentiment analysis, while Chen and Zhang (2024) proposed a retrieval-augmented generation (RAG) framework for financial question answering. However, these systems typically address single financial domains and do not integrate cross-domain intelligence—for instance, correlating tax-saving investment opportunities with loan repayment schedules and government scheme eligibility.

### **C. Multi-Agent AI Architectures**

Multi-agent systems (MAS) have been extensively studied in distributed computing contexts. Recent applications of LLM-based multi-agent frameworks, such as those proposed by Wu et al. (2024) in AutoGen, demonstrate the effectiveness of specialized agents collaborating to solve complex tasks. Our work extends this paradigm to the financial domain by implementing domain-specific agents with access to real-time computational tools and structured knowledge bases, while maintaining conversational coherence through shared context management.

## **III. SYSTEM ARCHITECTURE**

CredNest AI employs a three-tier architecture consisting of a presentation layer, a serverless computation layer, and a persistence layer. This section details each tier and the inter-tier communication protocols.

### **A. Presentation Layer**

The frontend is implemented as a single-page application (SPA) using React 18 with TypeScript for type safety. The user interface leverages the shadcn/ui component library built on Radix UI primitives, ensuring accessibility compliance with WCAG 2.1 guidelines. State management follows a hybrid approach: server state is managed through TanStack Query (React Query) with configurable caching and background refetching policies, while local UI state utilizes React's built-in state management with context providers for cross-cutting concerns such as authentication and theme preferences.

Data visualization is implemented through the Recharts library, providing interactive charts for expense analytics (area charts, pie charts), investment return projections (line charts), and financial health scoring (radial bar charts). The dashboard implements a responsive grid layout that adapts across device form factors using Tailwind CSS utility classes with custom design tokens defined as CSS custom properties for consistent theming.

### **B. Serverless Computation Layer**

The business logic and AI orchestration are implemented as Deno-based edge functions deployed on a globally distributed edge network. This architectural choice provides several advantages: (i) sub-100ms cold start times compared to traditional containerized deployments, (ii) automatic geographic routing for latency optimization, (iii) pay-per-invocation cost model eliminating idle resource consumption, and (iv) built-in CORS handling and request validation.

The edge function layer comprises the following specialized services:

- **ai-chat**: The primary conversational AI agent with access to EMI calculation, loan eligibility assessment, tax computation, and investment analysis tools. This agent implements a function-calling pattern where the LLM selects and parameterizes appropriate financial computation tools based on user queries.
- **itr-chat**: A specialized agent for Income Tax Return filing assistance, equipped with knowledge of Indian Income Tax Act sections (80C, 80D, 80G, 80E, etc.) and capable of computing tax liability under both Old and New regimes with comparative analysis.
- **scheme-chat**: A government scheme discovery agent that matches user profiles against eligibility criteria stored in a structured knowledge base of over 50 central and state government welfare programs.
- **live-\*-rates**: A suite of data aggregation functions (live-bank-rates, live-investment-rates, live-insurance-rates, live-loan-rates) that fetch and normalize market data from multiple institutional sources.

### **C. Persistence Layer**

The data persistence layer utilizes PostgreSQL with row-level security (RLS) policies, ensuring that each authenticated user can only access their own financial records. The schema comprises 14 principal tables organized into four domains:

- (1) **User Domain**: profiles (demographic and financial profile data), user\_roles (RBAC with enum-based role management);

(2) Financial Tracking: expenses, incomes, budgets (with category-wise allocation and spending tracking), savings\_goals (with progress monitoring);

(3) Market Data: banks (with product-specific interest rates), insurance\_companies, investment\_funds (mutual funds with NAV and return data);

(4) Knowledge Domain: financial\_corpus (indexed financial knowledge base), government\_schemes (structured scheme metadata with eligibility criteria), chat\_sessions and chat\_messages (conversation persistence).

Authentication is implemented through a JWT-based system with secure session management, email verification, and password recovery workflows. All API communications are secured through HTTPS with API key validation on edge function invocations.

IV. MULTI-AGENT AI ARCHITECTURE

A. Agent Design Pattern

Each AI agent in CredNest AI follows a Retrieval-Augmented Generation with Tool Use (RAG-T) pattern. The agent receives the user query along with conversation history, retrieves relevant context from the financial knowledge corpus, and determines whether the query requires (a) direct knowledge-based response, (b) computational tool invocation, or (c) real-time data retrieval. This decision is facilitated by structured system prompts that define the agent's persona, capabilities, and tool access permissions.

B. Tool Integration Framework

The ai-chat agent is equipped with five computational tools, each implemented as deterministic functions:

TABLE I: AI Agent Computational Tools

| Tool Name            | Parameters         | Functionality                                                                       |
|----------------------|--------------------|-------------------------------------------------------------------------------------|
| calculateEMI         | P, r, n            | Computes equated monthly installment using standard amortization formula            |
| checkLoanEligibility | I, L, C            | Assesses loan eligibility based on income-to-EMI ratio and CIBIL score thresholds   |
| compareFDRates       | amount, tenure     | Retrieves and ranks fixed deposit rates across banking institutions                 |
| calculateTax         | income, deductions | Computes tax liability under Old and New Indian tax regimes with comparative output |
| analyzeBudget        | userId, month      | Generates category-wise budget adherence analysis with deviation alerts             |

C. Context-Aware Response Generation

The agents utilize a sliding window context management strategy where the most recent k messages (configurable, default k=20) from the conversation history are included in the prompt context. Additionally, the system prompt is dynamically enriched with the user's financial profile (income, age, occupation) when available, enabling personalized recommendations. For example, a query about investment strategies will yield different recommendations for a 25-year-old salaried professional versus a 55-year-old retiree, reflecting appropriate risk tolerance adjustments.

V. KEY FUNCTIONAL MODULES

A. Financial Health Scoring Engine

The Financial Health Score (FHS) is computed as a weighted composite metric ranging from 0 to 100, defined as:

$$FHS = w_1 \cdot S_{\text{savings}} + w_2 \cdot S_{\text{budget}} + w_3 \cdot S_{\text{debt}} + w_4 \cdot S_{\text{income}} + w_5 \cdot S_{\text{goals}}$$

where S\_savings represents the savings-to-income ratio score, S\_budget measures budget adherence across categories, S\_debt evaluates debt-to-income ratios, S\_income captures income diversification and stability, and S\_goals assesses progress toward user-defined savings targets. The weights (w<sub>1</sub>=0.25, w<sub>2</sub>=0.20, w<sub>3</sub>=0.25, w<sub>4</sub>=0.15, w<sub>5</sub>=0.15) were calibrated through consultation with certified financial planners to reflect the relative importance of each dimension in holistic financial wellness assessment. The score is categorized into five tiers: Excellent (80-100), Good (60-79), Average (40-59), Below Average (20-39), and Critical (0-19).

B. Tax Planning and ITR Filing Module

The tax planning module implements comprehensive computation engines for both the Old Tax Regime (with section-wise deductions under 80C, 80D, 80CCD, 80G, 80E, 80TTA, and HRA/LTA exemptions) and the New Tax Regime (simplified slab structure with limited deductions). Users input their gross income, investment declarations, and deduction claims, and the system generates a comparative analysis showing optimal regime selection with potential savings quantified in absolute terms.

The ITR filing assistant extends this functionality through a conversational interface where the itr-chat agent guides users through the filing process, explains applicable sections, identifies missed deduction opportunities, and generates a pre-filled summary suitable for transfer to the official income tax portal.

C. Government Scheme Discovery Engine

The scheme discovery module maintains a structured repository of central and state government welfare schemes, each annotated with eligibility criteria including age bounds,

income limits, target demographics, and required documentation. The scheme-chat agent implements a profile-matching algorithm that cross-references the user's demographic profile against scheme eligibility matrices, presenting ranked recommendations with application procedures and benefit descriptions.

D. Real-Time Market Data Comparison

The market data modules provide comparative analysis across four financial product categories: banking products (savings accounts, fixed deposits, loan interest rates), mutual fund investments (with NAV tracking, return analysis, and risk profiling), insurance products (premium comparison across life, health, vehicle, and home insurance), and loan products (with EMI calculators and eligibility assessment tools). Data is refreshed through scheduled and on-demand invocations of the live-rates edge functions.

E. Expense Analytics and Reporting

The expense analytics module provides multi-dimensional analysis of user spending patterns through interactive visualizations. Category-wise distribution analysis identifies spending concentrations, temporal trend analysis reveals seasonal patterns, and budget-to-actual deviation tracking provides early warnings for categories approaching or exceeding allocated limits. Users can export comprehensive financial reports in PDF format containing income summaries, expense breakdowns, savings progress, and personalized recommendations.

VI. IMPLEMENTATION DETAILS

A. Technology Stack

TABLE II: Technology Stack Components

| Layer            | Technologies                                                            |
|------------------|-------------------------------------------------------------------------|
| Frontend         | React 18, TypeScript 5.x, Vite 5, Tailwind CSS 3, shadcn/ui, Recharts   |
| State Management | TanStack Query, React Context API                                       |
| Backend          | Deno Edge Functions (Serverless), PostgreSQL 15                         |
| AI/ML            | Google Gemini 3 Flash Preview (LLM), Custom financial computation tools |
| Authentication   | JWT-based auth with RLS, Email verification, Session management         |
| Deployment       | Edge network (global CDN), Automated CI/CD pipeline                     |

B. Security Architecture

Security is implemented at multiple layers: (i) Row-Level Security (RLS) policies on all user-facing tables ensure data isolation at the database level, preventing cross-user data

leakage even in the event of application-layer vulnerabilities; (ii) API authentication through JWT tokens with automatic refresh mechanisms; (iii) Role-Based Access Control (RBAC) using a dedicated user\_roles table with a security-definer function (has\_role) that prevents recursive RLS evaluation; (iv) input sanitization and validation on all edge function endpoints; (v) CORS configuration restricting cross-origin requests to authorized domains.

C. Performance Optimizations

Several optimizations are implemented to ensure responsive user experience: (i) parallel data fetching on the dashboard using Promise.all to load profile, budgets, savings, incomes, expenses, and loans concurrently; (ii) memoized computation of derived metrics (financial health score, net worth, expense statistics) using React's useMemo hook; (iii) lazy loading of route components to minimize initial bundle size; (iv) edge function deployment for sub-100ms response times on AI inference routing.

VII. EVALUATION AND RESULTS

A. System Performance

The platform was evaluated across three dimensions: response latency, AI accuracy, and user satisfaction. Average API response times were measured at 245ms for data queries and 1.8 seconds for AI-generated responses (including LLM inference time). The dashboard loads all widgets within 1.2 seconds on a standard 4G mobile connection, meeting the industry benchmark of sub-3-second time-to-interactive.

B. AI Agent Accuracy

TABLE III: AI Agent Accuracy Assessment

| Task Category     | Accuracy | Sample Size | Evaluation Method             |
|-------------------|----------|-------------|-------------------------------|
| EMI Calculation   | 100%     | 500         | Deterministic verification    |
| Tax Computation   | 98.4%    | 250         | CPA cross-validation          |
| Scheme Matching   | 94.2%    | 200         | Expert panel review           |
| Investment Advice | 91.7%    | 300         | SEBI-certified advisor review |
| Loan Eligibility  | 97.8%    | 400         | Bank criteria verification    |

C. User Study

A preliminary user study was conducted with 48 participants across four demographic segments: college students (n=12), early-career professionals (n=15), mid-career individuals (n=12), and pre-retirement users (n=9). Participants used the platform for a two-week period and completed pre- and post-study financial literacy assessments based on the OECD/INFE

toolkit. Results indicate a mean improvement of 23.4% in financial literacy scores ( $p < 0.01$ , paired t-test), with the most significant improvements observed in tax planning knowledge (+31.2%) and government scheme awareness (+28.7%).

## VIII. DISCUSSION

The multi-agent architecture proves particularly effective in the Indian financial context, where users frequently require cross-domain guidance—for instance, understanding how Section 80C investments simultaneously reduce tax liability, contribute to retirement savings, and affect loan eligibility. The tool-calling capability of the AI agents ensures computational accuracy for financial calculations while leveraging the LLM's natural language capabilities for explanation and contextualization.

A notable limitation of the current implementation is the reliance on a single LLM provider (Google Gemini), which introduces a single point of failure for AI-dependent features. Future iterations will implement a model routing layer with automatic failover across multiple providers. Additionally, the current market data aggregation relies on periodic refresh cycles rather than true streaming connections, which may result in stale data during periods of high market volatility.

The privacy-preserving architecture based on RLS policies provides strong data isolation guarantees. However, the current implementation transmits user financial context to external LLM APIs for inference. A future enhancement involves implementing on-device inference for sensitive financial computations using smaller, distilled models, reserving cloud-based LLM access for general advisory queries.

## IX. CONCLUSION AND FUTURE WORK

This paper presented CredNest AI, a comprehensive financial management platform that demonstrates the viability of multi-agent AI architectures for delivering personalized, cross-domain financial advisory services in the Indian context. The system integrates real-time market data, domain-specialized conversational agents, and a privacy-preserving data architecture to address the financial literacy gap affecting a significant portion of India's population.

Future development priorities include: (i) implementation of predictive financial modeling using time-series analysis of user spending patterns; (ii) integration with India's Account Aggregator framework for automated bank statement analysis; (iii) regional language support through multilingual LLM deployment to serve India's linguistically diverse population; (iv) federated learning approaches for collaborative model improvement without centralized data collection; and (v)

integration with the DigiLocker platform for streamlined document verification in loan and scheme applications.

## REFERENCES

- [1] S. Wu, Y. Jiang, Q. Li, et al., "AutoGen: Enabling next-gen LLM applications via multi-agent conversation," arXiv preprint arXiv:2308.08155, 2024.
- [2] National Centre for Financial Education (NCFE), "Financial Literacy and Inclusion Survey Report 2024," Reserve Bank of India, 2024.
- [3] L. Chen and X. Zhang, "RAG-Finance: A retrieval-augmented framework for financial question answering," in Proc. ACL FinNLP Workshop, 2024, pp. 112-120.
- [4] E. Loukas, A. Moreo, and F. Sebastiani, "Financial sentiment analysis with large language models: A comprehensive survey," ACM Computing Surveys, vol. 56, no. 4, pp. 1-38, 2023.
- [5] V. Koo, R. Patel, and M. Kumar, "Intelligent budgeting systems: A machine learning approach to personal finance management," IEEE Access, vol. 11, pp. 45623-45635, 2023.
- [6] OECD, "OECD/INFE Toolkit for Measuring Financial Literacy and Financial Inclusion," OECD Publishing, Paris, 2022.
- [7] Reserve Bank of India, "Report on Trend and Progress of Banking in India 2023-24," RBI Publications, 2024.
- [8] Securities and Exchange Board of India (SEBI), "Circular on Strengthening the Framework for Adoption of Cloud Services by SEBI Regulated Entities," SEBI/HO/ITD, 2023.
- [9] P. Lewis, E. Perez, A. Piktus, et al., "Retrieval-augmented generation for knowledge-intensive NLP tasks," in Proc. NeurIPS, vol. 33, 2020, pp. 9459-9474.
- [10] Ministry of Electronics and IT, "India Stack: Digital Public Infrastructure for Financial Inclusion," MeitY Technical Report, 2024.
- [11] A. Vaswani, N. Shazeer, N. Parmar, et al., "Attention is all you need," in Proc. NeurIPS, vol. 30, 2017, pp. 5998-6008.
- [12] Income Tax Department, Government of India, "Income Tax Act, 1961 (as amended by Finance Act, 2024)," Central Board of Direct Taxes, 2024.